The Climate Problem: Compounding climate shocks—recurring multi-year droughts, severe groundwater reserve depletion (reaching ~60% in northeastern basins), extreme heatwaves, and escalating agricultural and forest-interface wildfires—threaten Syria's entire agricultural sector. Across Syria's diverse agro-ecological zones, growers lack localized environmental telemetry to guide irrigation, fire defense, and harvest timing. This causes wasted diesel and groundwater in irrigated plains, severe stress and yield loss in rain-fed orchards, devastating fires across both grain fields and perennial groves, and quality degradation from mistimed harvests.
Geography: All major agricultural zones across Syria—encompassing the northern and eastern cereal plains (Al-Hasakah, Raqqa, Deir ez-Zor), central irrigated basins (Al-Ghab, Hama, Homs), southern agricultural hubs (Daraa, Sweida), and coastal/western orchard belts (Latakia, Tartus, Idlib, Aleppo).
Intended User (Customer Segment): All Syrian farmers and growers—from smallholder family farms to medium-to-large commercial estates, agricultural cooperatives, and local agronomic NGOs/extension networks—cultivating annual field crops (wheat, barley, cotton, legumes, vegetables) as well as perennial fruit and nut trees (olives, citrus, pistachios, apples, grapes, stone fruits).

2. SDG Pathway
Primary — SDG 13 (Climate Action — Target 13.1): Strengthen the climate resilience and adaptive capacity of all Syrian farmers against droughts, heatwaves, and wildfire hazards through localized early warning systems and actionable tactical guidance.
Secondary — SDG 6 (Clean Water and Sanitation — Target 6.4): Substantially increase agricultural water-use efficiency and curb unsustainable groundwater extraction across diverse crop types through precision deficit-irrigation scheduling.
Co-Benefit — SDG 2 (Zero Hunger — Target 2.4): Safeguard national food security and rural livelihoods by protecting both staple annual crops and high-value perennial orchards from extreme weather and fire loss.

3. Proposed Technical Approach
Built in **Python** using **Machine Learning** and geospatial remote sensing pipelines, the platform acts as a hardware-free "virtual agronomist" tailored to different crop phenologies (annuals vs. perennial orchards):
Crop-Aware Drought & Irrigation Early Warning: Automates ingestion of multispectral satellite data and weather forecasts (`geopandas`, `rasterio`, Google Earth Engine API). A crop-stratified Gradient Boosting model (XGBoost/LightGBM) combines vegetation/water indices (NDVI, NDWI, thermal LST) with reference evapotranspiration (ET0ET_0) and crop coefficients (KcK_c) to forecast 7-to-14-day root-zone water stress. It prescribes tailored irrigation volumes for field crops and strategic survival/deficit-irrigation schedules for drought-stressed olive, citrus, and pistachio orchards when water or fuel is rationed.
Field & Orchard Wildfire Early Warning: Cross-references near-real-time thermal anomalies with vegetation dryness indices (NDMI/NBR) and wind-vector forecasts. Spatial proximity algorithms trigger immediate alerts when upwind hotspots threaten farm fields or coastal/mountainous orchards, instructing farmers on preventive firebreak

ploughing, brush clearing, and perimeter wetting.
Crop-Specific Optimal Harvest Timing: Applies time-series curve fitting to satellite phenology trajectories (senescence curves for grains; canopy vigor and thermal growing-degree days for orchards) alongside 10-day weather and fire-risk forecasts to pinpoint optimal harvest windows—protecting grains from shatter/fire and fruits/olives from heatwaves, frost, or unseasonal rain.
Accessible Low-Bandwidth Delivery: Translates ML outputs via a Python rule-based/LLM instruction engine into clear, localized Arabic tactical directives delivered over WhatsApp, Telegram, or SMS—requiring zero hardware installation for smallholders.

4. Candidate Data Sources (Open-Access)
Multispectral & Thermal Satellite Imagery: ESA Sentinel-2 (10m resolution, 5-day revisit—fine enough to monitor both open fields and individual orchard blocks for NDVI, NDWI, NDRE, and fuel dryness) and **NASA/USGS Landsat 8 & 9** (Land Surface Temperature and evapotranspiration stress).
Near-Real-Time Active Fire & Thermal Data: NASA FIRMS (VIIRS 375m and MODIS active fire/hotspot detections) for rapid wildfire proximity alerts across plains and coastal/mountain orchard belts.
Meteorological & Forecast Data: ERA5-Land (Copernicus) and **NASA POWER** for precipitation, wind vectors, relative humidity, growing-degree days, and ET0ET_0 calculations.
Baseline Calibration & Land-Cover Data: FAO WaPOR (Middle East water productivity portal) and **ESA WorldCover / FAO GAEZ** for crop-type masks, perennial tree-cover differentiation, and regional water-use benchmarks.

5. Relevant References
FAO (2026). *GIEWS Country Brief: Syrian Arab Republic (August 2026) & Emergency and Resilience Plan 2026–2028*. Food and Agriculture Organization of the United Nations. *(Documents nationwide climate impacts on both cereal harvests and perennial tree crops, water scarcity, and agricultural resilience priorities across Syria).*
Carnegie Endowment for International Peace / Sada (2026). *Syria on the Brink of Water Scarcity: Climate Change, Drought, and Threats to Food Security*. *(Details severe groundwater depletion across Syria's agricultural basins due to recurring droughts and its direct threat to farming livelihoods and food production).*